

Assignment 2 Report - ID3

Assignment 2 Data Mining Report - ID3

Program Inputs

- Classifier Choice: ID3
- Live Terminal Output: Yes
- T1 Threshold: 0.8
- g Parameter: 0.01
- Training File: A2--ID3-Training set.csv
- Test File: A2--ID3-TEST set.csv

What This Method Does

ID3 builds a decision tree from the training set, extracts readable IF-THEN rules, and uses those rules to classify the test set.

The pre-pruning threshold T1 controls early stopping during tree construction. The post-pruning parameter g controls whether a completed subtree can be removed after testing the effect on classification quality.

Main Result

After post-pruning, ID3 correctly classified 47.73% of the ID3 test records.

Formula Notes

- ID3 entropy/MC: $-\sum(p * \log_2(p))$.
- ID3 Gain: $MC(D) - WMC(\text{attribute})$.
- ID3 pre-pruning mixture ratio: $\alpha = c1 / |F1|$.
- ID3 post-pruning: if $(N - M) / Q \leq gK$, then the subtree can be removed.
- Naive Bayes prior: $\text{count}(\text{Volume class}) / \text{total training records}$.
- Naive Bayes categorical probability: $\text{count}(\text{attribute value and class}) / \text{count}(\text{class})$, with Laplace smoothing if needed.
- Naive Bayes continuous probability uses the Gaussian density formula with mean and sample standard deviation by class.

Output Tables Created

- ID3 Class Distribution
- ID3 Gain Calculations
- ID3 Pre-Pruning Decisions
- ID3 Post-Pruning Decisions
- ID3 Final Metrics